Questions

Q1.

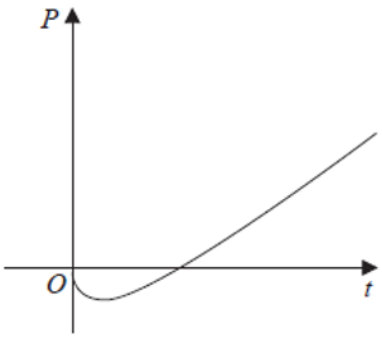


Figure 2

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The profit made by a company, £ P million, t years after the company started trading, is modelled by the equation

$$P = \frac{4t-1}{10} + \frac{3}{4}\ln\left[\frac{t+1}{(2t+1)^2}\right]$$

The graph of P against t is shown in Figure 2.

According to the model,

(a) show that exactly one year after it started trading, the company had made a loss of approximately £ 830 000

(2)

A manager of the company wants to know the value of t for which $P = 0$

(b) Show that this value of t occurs in the interval [6, 7]

(2)

(c) Show that the equation $P = 0$ can be expressed in the form

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$$t = \frac{1}{4} + \frac{15}{8}\ln\left[\frac{(2t+1)^2}{t+1}\right]$$

(2)

(d) Using the iteration formula

$$t_{n+1} = \frac{1}{4} + \frac{15}{8}\ln\left[\frac{(2t_n+1)^2}{t_n+1}\right] \quad \text{with} \quad t_1 = 6$$

find the value of t_2 and the value of t_6 , giving your answers to 3 decimal places.

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(3)

(e) Hence find, according to the model, how many months it takes in total, from when the company started trading, for it to make a profit.

(2)

(Total for question = 11 marks)

(Q05 WMA13/01, Oct 2022)

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Q2.

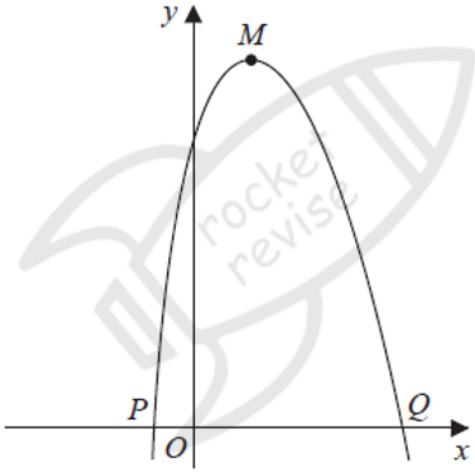


Figure 1

Figure 1 shows a sketch of part of the curve with equation

$$y = 6\ln(2x+3) - \frac{1}{2}x^2 + 4 \qquad x > -\frac{3}{2}$$

The curve cuts the negative x-axis at the point P, as shown in Figure 1.

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(a) Show that the x coordinate of P lies in the interval [–1.25, –1.2]

(2)

The curve cuts the positive x-axis at the point Q, also shown in Figure 1.

Using the iterative formula

$$x_{n+1} = \sqrt{12\ln(2x_n + 3) + 8} \quad \text{with} \quad x_1 = 6$$

(b) (i) find, to 4 decimal places, the value of x^2

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(ii) find, by continued iteration, the x coordinate of Q. Give your answer to 4 decimal places.

(3)

The curve has a maximum turning point at M, as shown in Figure 1.

(c) Using calculus and showing each stage of your working, find the x coordinate of M.

(4)

(Total for question = 9 marks)

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(Q05 WMA13/01, Jan 2022)

Q3.

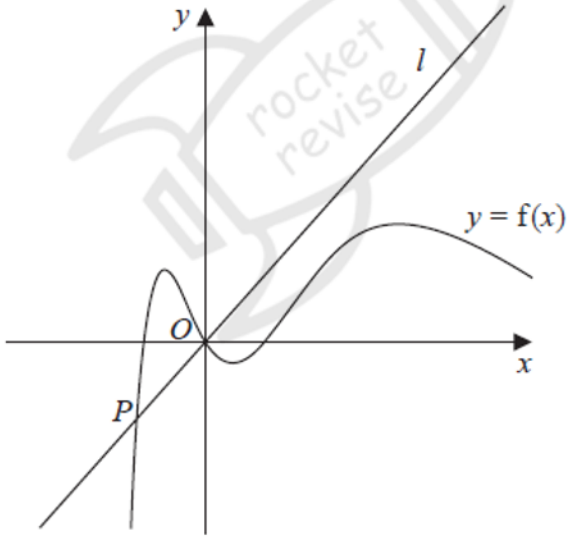


Figure 3

Figure 3 shows a sketch of part of the curve with equation $y = f(x)$, where

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$$f(x) = x(x^2 - 4)e^{-\frac{1}{2}x}$$

(a) Find $f'(x)$.

(2)

The line l is the normal to the curve at O and meets the curve again at the point P.

The point P lies in the 3rd quadrant, as shown in Figure 3.

(b) Show that the x coordinate of P is a solution of the equation

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$$x = -\frac{1}{2}\sqrt{16 + e^{\frac{1}{2}x}}$$

(4)

(c) Using the iterative formula

$$x_{n+1} = -\frac{1}{2}\sqrt{16 + e^{\frac{1}{2}x_n}} \quad \text{with } x_1 = -2$$

find, to 4 decimal places,

(i) the value of x_2

(ii) the x coordinate of P.

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(3)

(Total for question = 9 marks)

(Q09 WMA13/01, Oct 2021)

Q4.

$$g(x) = x^6 + 2x - 1000$$

(a) Show that $g(x) = 0$ has a root a in the interval [3, 4]

(2)

Using the iteration formula

$$x_{n+1} = \sqrt[6]{1000 - 2x_n} \quad \text{with } x_1 = 3$$

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(b) (i) find, to 4 decimal places, the value of x^2

(ii) find, by repeated iteration, the value of a.

Give your answer to 4 decimal places.

(3)

(Total for question = 5 marks)

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(**Q01 WMA13/01, June 2023**)

Q5.

$$f(x) = x^3 + 3x^2 + 4x - 12$$

(a) Show that the equation $f(x) = 0$ can be written as

$$x = \sqrt{\left(\frac{4(3-x)}{(3+x)}\right)} \quad x \neq -3$$

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(3)

The equation $x^3 + 3x^2 + 4x - 12 = 0$ has a single root which is between 1 and 2

(b) Use the iteration formula

$$x_{n+1} = \sqrt{\left(\frac{4(3-x_n)}{(3+x_n)}\right)} \quad n \geqslant 0$$

with $x_0 = 1$ to find, to 2 decimal places, the value of x_1, x_2 and x_3

(3)

The root of $f(x) = 0$ is α.

(c) By choosing a suitable interval, prove that $\alpha = 1.272$ to 3 decimal places.

(2)

(Total for question = 8 marks)

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(**Q02 WMA13/01, Specimen papers**)

Q6.

A curve has equation $y = f(x)$ where

$$f(x) = x^2 - 5x + e^x \qquad x \in \mathbb{R}$$

(a) Show that the equation $f(x) = 0$ has a root, α, in the interval [1, 2]

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(2)

The iterative formula

$$x_{n+1} = \sqrt{5x_n - e^{x_n}}$$

with $x^1 = 1$ is used to find an approximate value for the root α.

(b) (i) Find the value of x_2 to 4 decimal places.

(ii) Find, by repeated iteration, the value of α, giving your answer to 4 decimal places.

(3)

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(Total for question = 5 marks)

(Q01 WMA13/01, Oct 2023)

Q7.

$$f(x) = x\cos\left(\frac{x}{3}\right) \quad x > 0$$

(a) Find f′(x)

(2)

(b) Show that the equation f′(x) = 0 can be written as

$$x = k\arctan\left(\frac{k}{x}\right)$$

where k is an integer to be found.

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(2)

(c) Starting with $x_1 = 2.5$ use the iteration formula

$$x_{n+1} = k\arctan\left(\frac{k}{x_n}\right)$$

with the value of k found in part (b), to calculate the values of x_2 and x_6 giving your answers to 3 decimal places.

(2)

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(d) Using a suitable interval and a suitable function that should be stated, show that a root of f'(x) = 0 is 2.581 correct to 3 decimal places.

(2)

(Total for question = 8 marks)

(Q06 WMA13/01, Jan 2021)

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Q8.

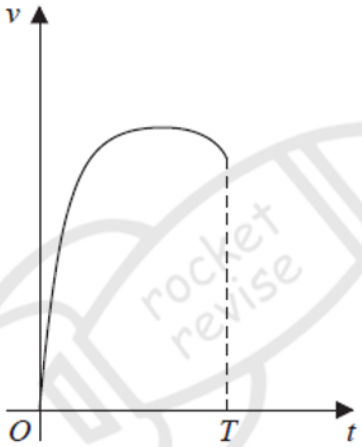


Figure 4

Figure 4 is a graph showing the velocity of a sprinter during a 100 m race.

The sprinter's velocity during the race, v m s^{-1}, is modelled by the equation

$$v = 12 - e^{t-10} - 12e^{-0.75t} \qquad t \geq 0$$

where t seconds is the time after the sprinter begins to run.

According to the model,

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(a) find, using calculus, the sprinter's maximum velocity during the race.

(5)

Given that the sprinter runs 100 m in T seconds, such that

$$\int_0^T v \, dt = 100$$

(b) show that T is a solution of the equation

$$T =$$

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$\frac{1}{12}(116 - 16e^{-0.75T} + e^{T-10} - e^{-10})$

(4)

The iteration formula

$$T_{n+1} = \frac{1}{12}(116 - 16e^{-0.75T_n} + e^{T_n-10} - e^{-10})$$

is used to find an approximate value for T

Using this iteration formula with $T_1 = 10$

(c) find, to 4 decimal places,

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(i) the value of T_2

(ii) the time taken by the sprinter to run the race, according to the model.

(3)

(Total for question = 12 marks)

(Q08 WMA13/01, June 2022)

Q9.

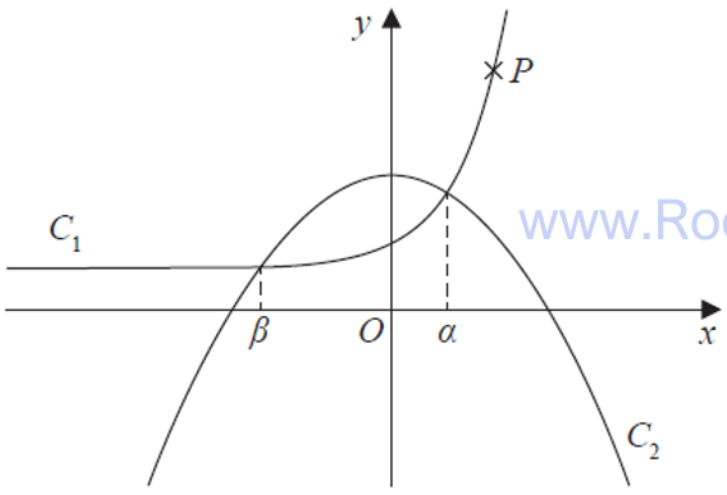


Figure 3

Figure 3 shows a sketch of curve C_1 with equation $y = 5e^{x-1} + 3$

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and of curve C_2 with equation $y = 10 - x^2$

The point P lies on C_1 and has y coordinate 18

(a) Find the x coordinate of P, writing your answer in the form $\ln k$, where k is a constant to be found.

(3)

The curve C_1 meets the curve C_2 at $x = \alpha$ and at $x = \beta$, as shown in Figure 3.

(b) Using a suitable interval and a suitable function that should be stated, show that to 3 decimal places $\alpha = 1.134$

(3)

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The iterative equation

$$x_{n+1} = -\sqrt{7 - 5e^{x_n - 1}}$$

is used to find an approximation to β.

Using this iterative formula with $x_1 = -3$

(c) find the value of x_2 and the value of β, giving each answer to 6 decimal places.

(3)

(Total for question = 9 marks)

(**Q06 WMA13/01, Oct 2020**)

Q10.

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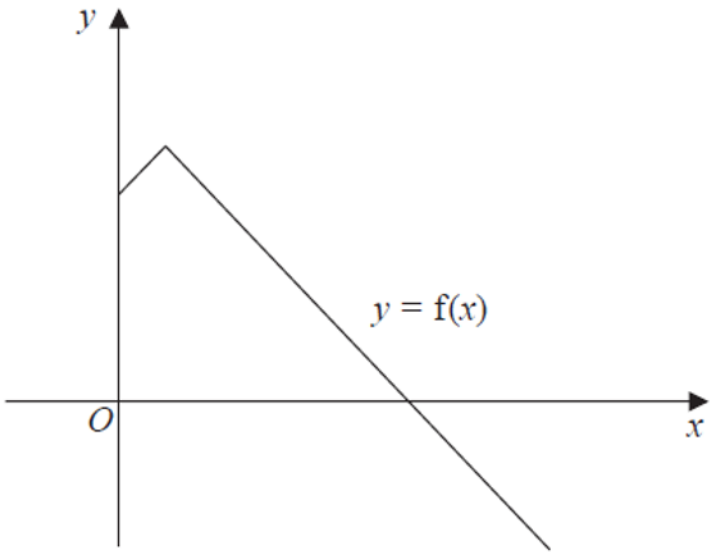


Figure 2

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Figure 2 shows a sketch of part of the graph with equation y = f(x) where

$$f(x) = 21 - 2|2 - x| \qquad x \geqslant 0$$

(a) Find ff(6)

(2)

(b) Solve the equation f(x) = 5x

(2)

Given that the equation f(x) = k , where k is a constant, has exactly two roots,

(c) state the set of possible values of k.

(2)

The graph with equation y = f(x) is transformed onto the graph with equation $y = a\,f(x - b)$

The vertex of the graph with equation $y = a\,f(x - b)$ is (6, 3).

Given that a and b are constants,

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(d) find the value of a and the value of b.

(2)

(Total for question = 8 marks)

(Q04 WMA13/01, Oct 2020)

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Q11.

The curve C has equation

$$y = x^2 \cos\left(\frac{1}{2}x\right) \qquad 0 < x \leq \pi$$

The curve has a stationary point at the point P.

(a) Show, using calculus, that the x coordinate of P is a solution of the equation

$$x = 2 \arctan\left(\frac{4}{x}\right)$$

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(4)

Using the iteration formula

$$x_{n+1} = 2 \arctan\left(\frac{4}{x_n}\right) \qquad x_1 = 2$$

(b) find the value of x_2 and the value of x_6, giving your answers to 3 decimal places.

(3)

(Total for question = 7 marks)

(Q01 WMA13/01, June 2021)

Q12.

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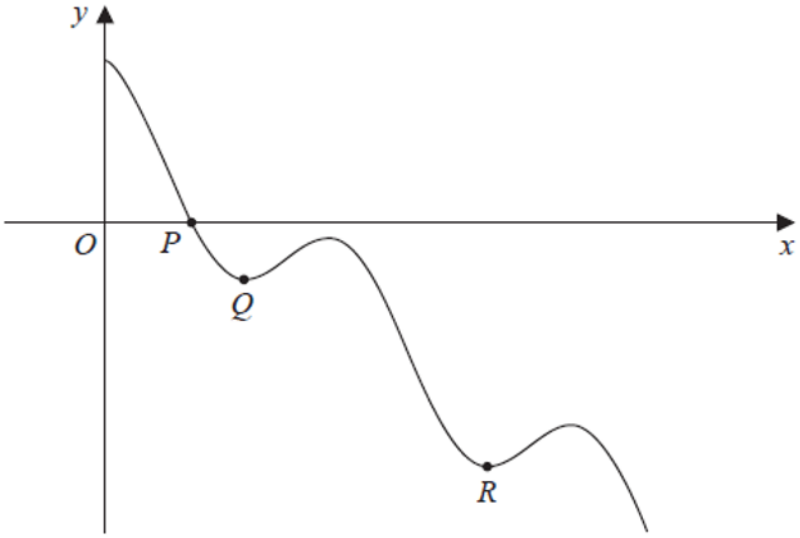


Figure 3

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Figure 3 shows a sketch of part of the curve with equation

$$y = 2\cos 3x - 3x + 4 \qquad x > 0$$

where x is measured in radians.

The curve crosses the x-axis at the point P, as shown in Figure 3.

Given that the x coordinate of P is α,

(a) show that α lies between 0.8 and 0.9

(2)

The iteration formula

$$x_{n+1} = \frac{1}{3}\arccos(1.5x_n - 2)$$

can be used to find an approximate value for α.

(b) Using this iteration formula with $x_1 = 0.8$ find, to 4 decimal places, the value of

(i) x_2

(ii) x_5

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(3)

The point Q and the point R are local minimum points on the curve, as shown in Figure 3.

Given that the x coordinates of Q and R are β and λ respectively, and that they are the two smallest values of x at which local minima occur,

(c) find, using calculus, the exact value of β and the exact value of λ.

(6)

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(Total for question = 11 marks)

(Q07 WMA13/01, Jan 2020)

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